

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Discrete Structure

Semester: Fall

Year : 2023
Full marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain Extended Euclidean algorithm with example. What is Inclusion-Exclusion Principle.
b) Define Injective and Bijective Functions. Show that the function $f: R \rightarrow R$, given by $f(x) = 2x + 7$, is one-one and onto.
2. a) Let $A = \{a, b, c, d\}$ and $R = \{(a, b), (b, c), (c, d), (b, a)\}$. Find the transitive close of R using Warshall's algorithm.
b) Define Equivalence and Partial order relation. Let R be the relation on the set of integers such that aRb if and only if "a-b" is even. Is R an equivalence relation?
3. a) Find the solution to the Recurrence Relation given by: $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ where $a_0 = 2$, $a_1 = 5$, and $a_2 = 15$.

OR

Explain Tower of Hanoi problem with its solution.

- b) What do you understand by Recurrence Relation? Explain the types of recurrence relation with examples.
4. a) Verify following argument is valid using the rules of inference: If you send me an e-mail message, then I will finish writing the program. If you do not send me an e-mail message, then I will go to sleep early. If I go to sleep early, then I will wake up feeling refreshed, leading to the conclusion that if I do not finish writing the program, then I will wake up feeling refreshed.
b) Consider the following statement. Express below statement using quantifiers where domain consist of all birds.
 - i. All humming birds are richly colored.
 - ii. No large birds lives on honey.

- iii. Birds that do not live on honey are dull in color.
- iv. Humming birds are small.

OR

Explain different types of quantifiers used in predicate logic.

5. a) Explain simple graph, Multigraph, Pseudo graph, with example.
b) What is the Euler's formula for planar graph? Prove the case where adding an edge, vertices remain constant but Face increases.
c) Differentiate between Hamilton graph and Euler's graph along with necessary diagrams.
6. a) What are the differences between DFA and N-DFA? Design DFA with $\Sigma = \{0,1\}$ accepts even number of 0's and even number of 1's.
b) Can we convert regular expression into finite automata? Explain with example.
7. Write short notes on: (Any two) [2 x 5]
 - a) Transport Network
 - b) Pigeonhole principle
 - c) Tautology and contradiction

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Attempt all the questions.

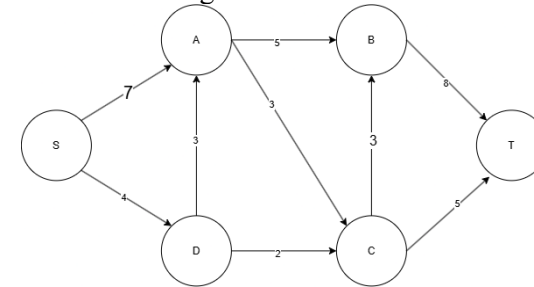
- Explain product and sum rule of counting. Show that if any eight positive integers are chosen two of them will have same remainder when divided by 7.
 - Define Relatively prime and Pairwise Relative prime numbers. Find the GCD of 414 and 662 using Euclidean Algorithm and express it as a linear combination of 414 and 662.

OR

Define Bijective Function. Prove that there are infinitely many prime numbers.

- Let $A = \{5, 6, 7, 8\}$ and $R = \{(5, 6), (6, 7), (7, 8), (6, 5)\}$. Find the Reflexive, Symmetric and Transitive closure of R.
 - Define POSET. Let R be the relation on the set of integers such that aRb if and only if “a-b” is divisible by 5. Is R an equivalence relation?
- Solve TOH puzzle using iterative approach.
 - Find the solution of recurrence relation: $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ with $a_0 = 2, a_1 = 5$ and $a_2 = 15$.
- Verify following argument is valid using the rules of inference: “If I get the job and work hard, then I’ll get promoted.” “If I get promoted, then I’ll be happy.” “I will not be happy.” Therefore, “Either I will not get job or I will not work hard”
 - Using proof by contrapositive show that if $n^3 + 2n + 1$ is odd then n is even.
- What is chromatic number? Determine chromatic number of bipartite graph, complete graph, cycle graph and wheel graph with necessary figures.

b) Find the Max flow in the given network:



OR

What is Hamilton graph? State and prove necessary condition for existence of Euler path but not circuit in a graph.

- Describe the working of finite state automate along with necessary diagram and its applications.
 - Give formal definitions of DFA and NFA. Construct a deterministic finite automata that accepts all those strings that ends with “babb” over $\Sigma = \{a, b\}$.
- Write short notes on: (Any two)
 - Types of Grammar
 - Walk, Path and Trail
 - Application of Graph Theory
 - Regular Expressions

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Attempt all the questions.

- Differentiate between Permutation and Combination. The 5 letters in the list L, M, N, O, and P are to rearrange so that L is the third letter in the list and M is not next to L. How many such rearrangement are there?
 - Illustrate pigeonhole principle with an example. Express GCD of 161 and 28, and express it as a linear combination of 161 and 28.
- What is relation? Explain transitive, reflexive and equivalence relation with example.

OR

What are the closure properties of relation? Explain with example.

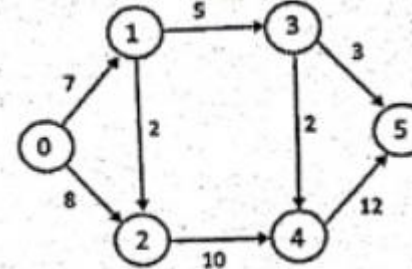
- Define equivalence relation. Is the relation, $\{(0,0), (1,1), (2,2), (3,3)\}$ on $\{0,1,2,3\}$ an equivalence relation? Justify your answer.
- Solve the following recurrence relation. $a_n = 5a_{n-1} - 6a_{n-2} + 42 \cdot 4^n$ with $a_0 = 56, a_1 = 278$.
- Find the explicit formula for Fibonacci sequence.

OR

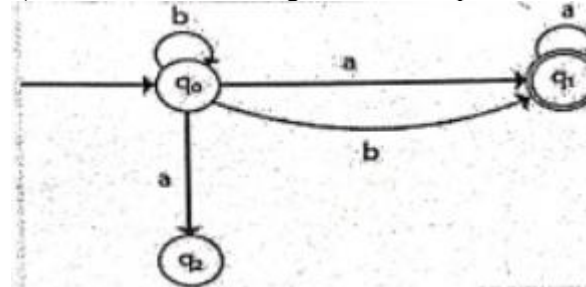
Solve TOH puzzle using iterative approach.

- Verify following argument is valid using the rules of inference: "If it rains, then the ground will be wet. If it does not rain, then the picnic will be held. If the picnic is held, then I will have a good time. Therefore, if the ground is not wet, then I will have a good time."
 - Prove the following using direct proof technique.
 - Sum of two odd numbers is even.
 - Square of odd numbers is also odd.
- Differentiate between Hamilton graph and Euler's graph along with necessary diagrams.

b) Find the max flow in the given network.



- Convert the following NFA to its equivalent DFA.



- Describe the working of finite state automata along with necessary diagram and its applications. Design a DFA that starts with 01 over alphabets $\{0, 1\}$.
- Write short notes on: (Any two)
 - Differentiate Walk, Path and Trail
 - Chomsky Hierarchy
 - Define Bipartite graph, and complete bipartite graph